

Практическая работа №7

ФИО: Смирнов Андрей Юрьевич

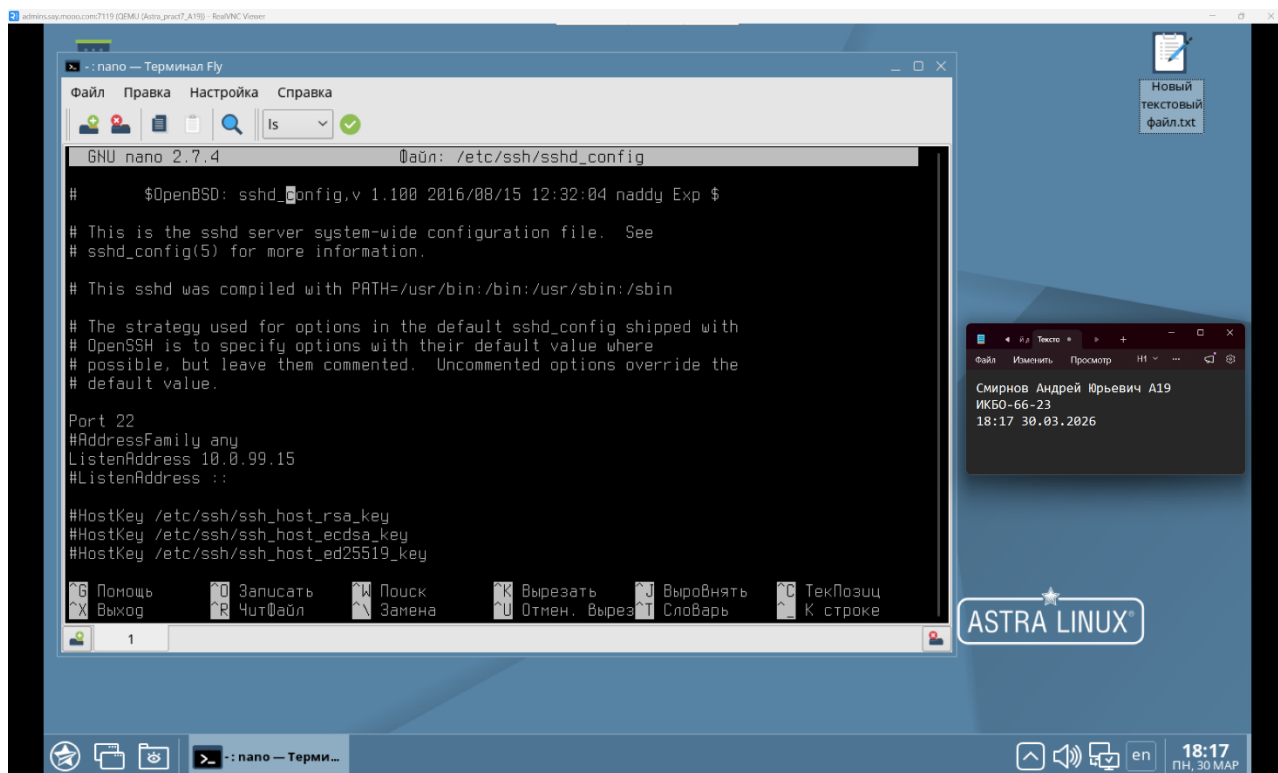
Группа: ИКБО-66-23

Данный отчет должен содержать скриншоты выполнения работы (замените скриншотом слово <..скриншот..> в соответствующем пункте).

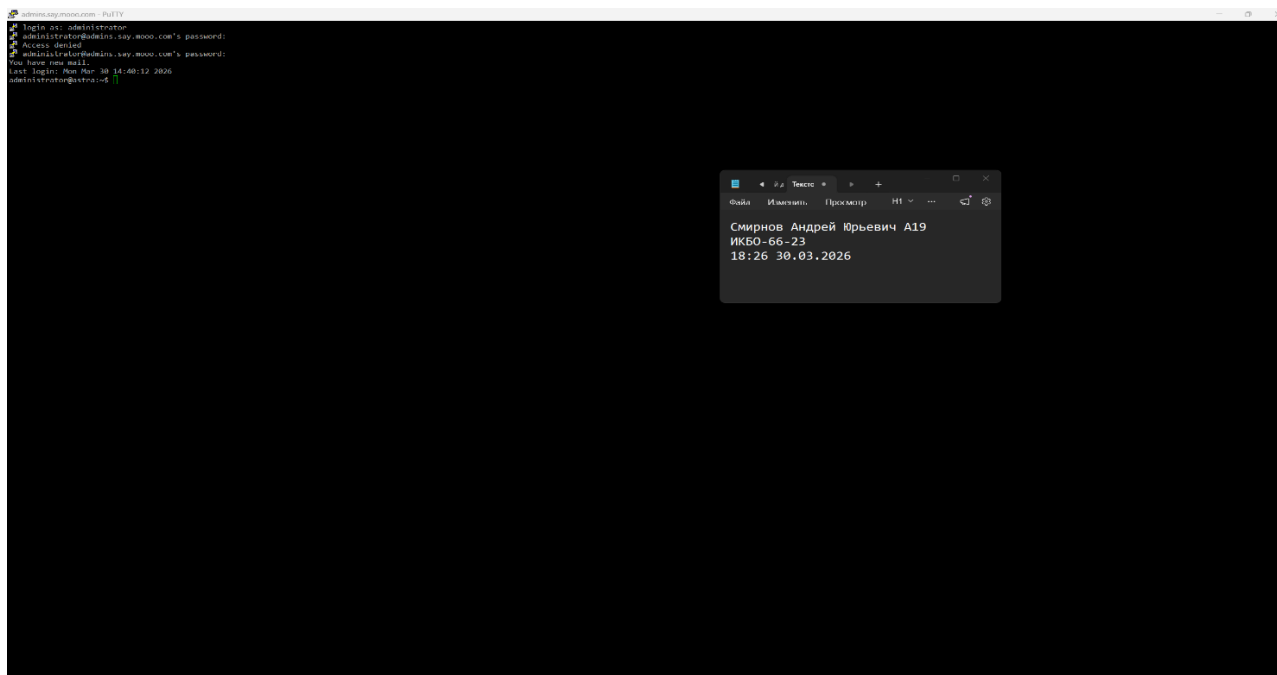
На **ВСЕХ** скриншотах, которые вы делаете, должно быть видно ваше ФИО и группу (для этого откройте блокнот и запишите их там), текущую дату и время и номер ВМ.

Задание 1: Настройка Astra Linux

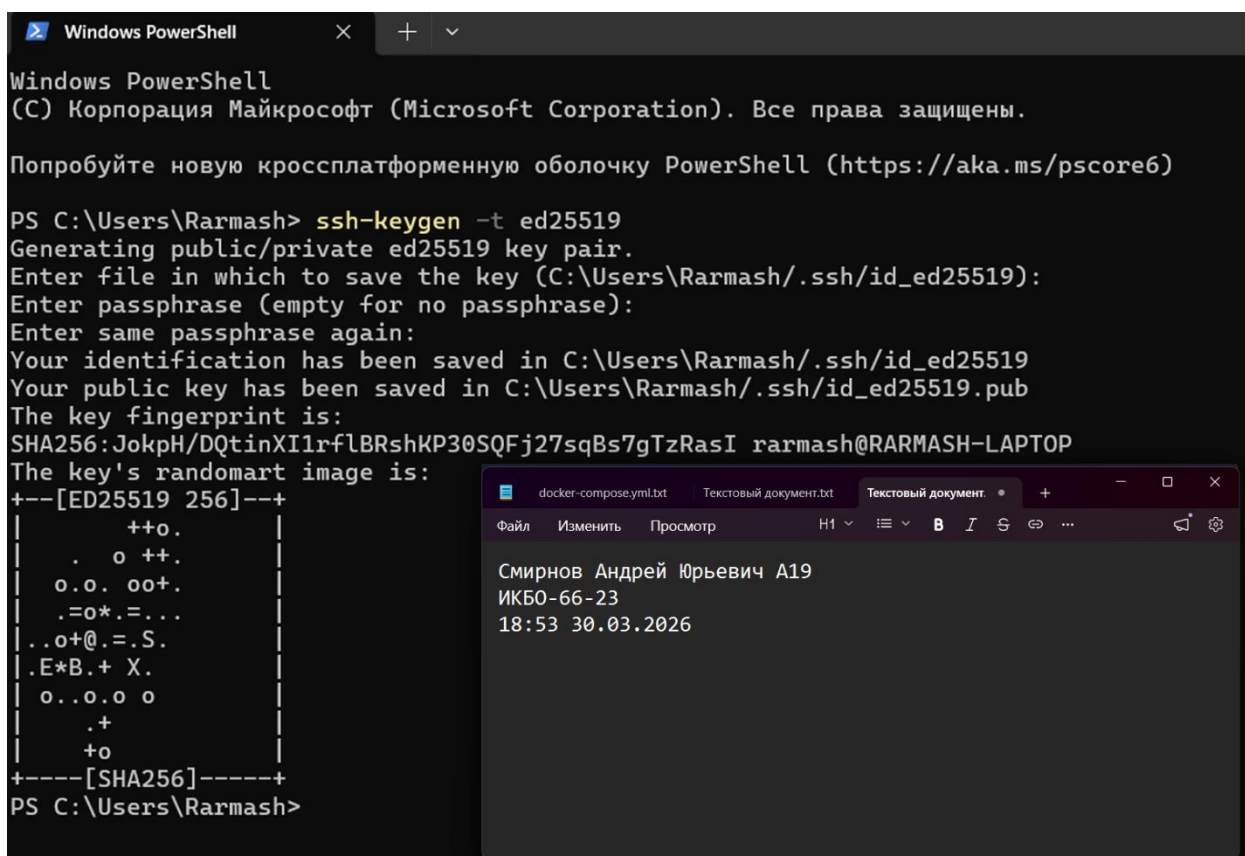
1. Конфигурация SSH



2. Подключение через PuTTY



3. Подключение через PuTTY без пароля с помощью ключа генерации



4. Результаты базовой инвентаризации компьютера

processor	: 0
vendor_id	: GenuineIntel
cpu family	: 6

model : 62
model name : Intel(R) Xeon(R) CPU E5-2697 v2 @ 2.70GHz
stepping : 4
microcode : 0x42d
cpu MHz : 2699.998
cache size : 16384 KB
physical id : 0
siblings : 1
core id : 0
cpu cores : 1
apicid : 0
initial apicid : 0
fpu : yes
fpu_exception : yes
cpuid level : 13
wp : yes
flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm
constant_tsc arch_perfmon rep_good nopl xtopology cpuid tsc_known_freq
pni pclmulqdq vmx ssse3 cx16 pdcm pcid sse4_1 sse4_2 x2apic popcnt
tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm cpuid_fault
pti ssbd ibrs ibpb stibp tpr_shadow vnmi flexpriority ept vpid fsgsbase
tsc_adjust smep erms xsaveopt arat umip arch_capabilities
vmx flags : vnmi preemption_timer posted_intr invvpid ept_x_only
ept_lgb flexpriority apicv tsc_offset vtptr mtf vapic ept vpid
unrestricted_guest vapic_reg vid shadow_vmcs
bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf
mds swapgs
bogomips : 5399.99
clflush size : 64

cache_alignment : 64

address sizes : 46 bits physical, 48 bits virtual

power management:

processor : 1

vendor_id : GenuineIntel

cpu family : 6

model : 62

model name : Intel(R) Xeon(R) CPU E5-2697 v2 @ 2.70GHz

stepping : 4

microcode : 0x42d

cpu MHz : 2699.998

cache size : 16384 KB

physical id : 1

siblings : 1

core id : 0

cpu cores : 1

apicid : 1

initial apicid : 1

fpu : yes

fpu_exception : yes

cpuid level : 13

wp : yes

flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov

pat pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm

constant_tsc arch_perfmon rep_good nopl xtopology cpuid tsc_known_freq

pni pclmulqdq vmx ssse3 cx16 pdcm pcid sse4_1 sse4_2 x2apic popcnt

tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm cpuid_fault

pti ssbd ibrs ibpb stibp tpr_shadow vnmi flexpriority ept vpid fsgsbase

tsc_adjust smep erms xsaveopt arat umip arch_capabilities

vmx flags : vnmi preemption_timer posted_intr invvpid ept_x_only
ept_lgb flexpriority apicv tsc_offset vtptr mtf vpic ept vpid
unrestricted_guest vpic_reg vid shadow_vmcs
bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf
mds swaps
bogomips : 5399.99
clflush size : 64
cache_alignment : 64
address sizes : 46 bits physical, 48 bits virtual
power management:

dmidecode 3.0

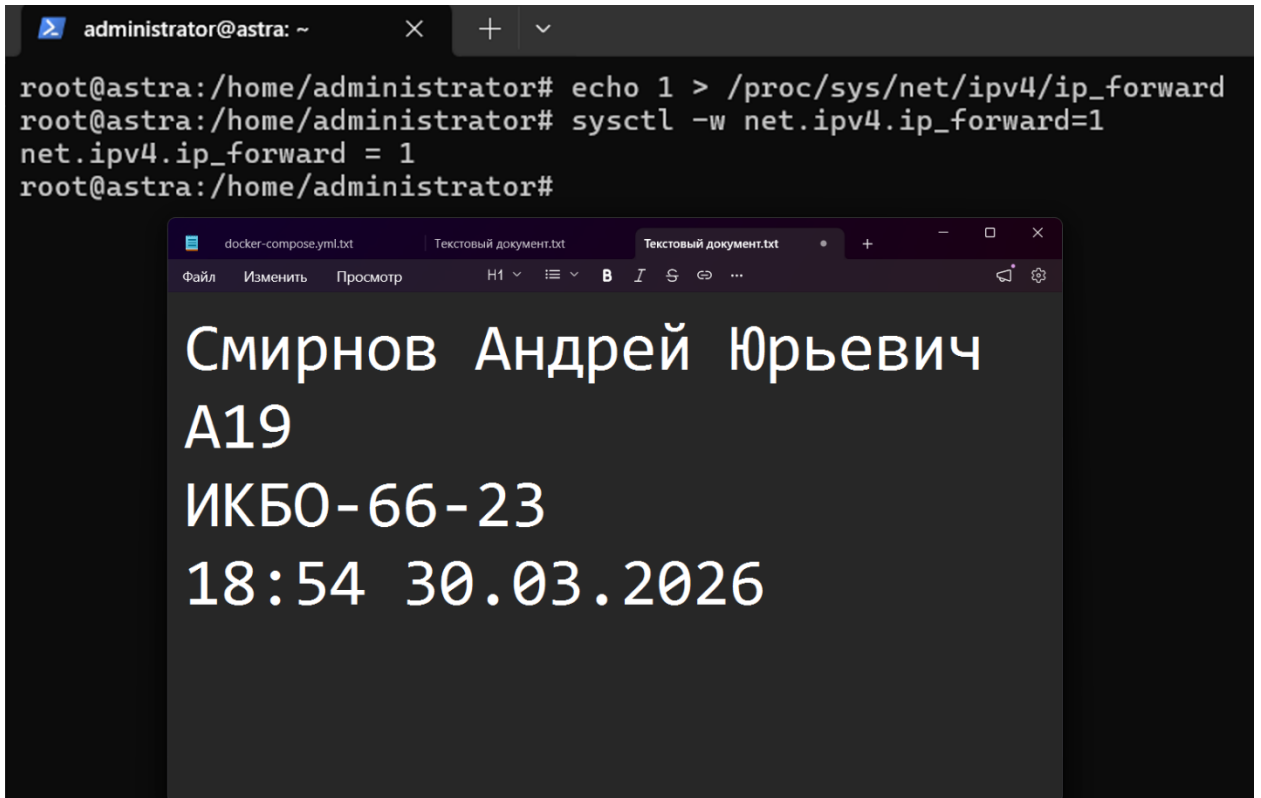
Getting SMBIOS data from sysfs.

SMBIOS 2.8 present.

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
pfifo_fast state UP group default qlen 1000
link/ether 52:54:00:46:91:a4 brd ff:ff:ff:ff:ff:ff
inet 10.0.99.15/24 brd 10.0.99.255 scope global eth0
valid_lft forever preferred_lft forever
inet6 fe80::5054:ff:fe46:91a4/64 scope link
valid_lft forever preferred_lft forever

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
pfifo_fast state UP group default qlen 1000
    link/ether 52:54:00:5c:8d:64 brd ff:ff:ff:ff:ff:ff
```

5. Включение перенаправления пакетов



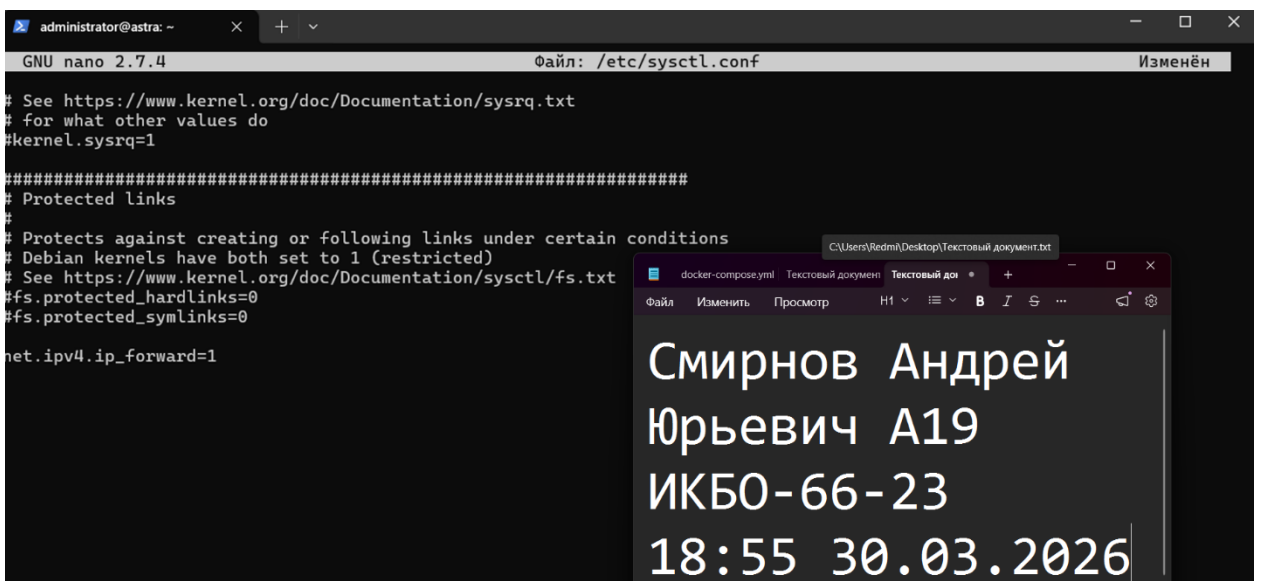
```
administrator@astra: ~
root@astra:/home/administrator# echo 1 > /proc/sys/net/ipv4/ip_forward
root@astra:/home/administrator# sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
root@astra:/home/administrator#
```

docker-compose.yml.txt Текстовый документ.txt Текстовый документ.txt

Файл Изменить Просмотр H1 B I S ...

Смирнов Андрей Юрьевич
A19
ИКБО-66-23
18:54 30.03.2026

6. Отредактированный файл конфигурации /etc/sysctl.conf



```
administrator@astra: ~
GNU nano 2.7.4    Файл: /etc/sysctl.conf    Изменён
# See https://www.kernel.org/doc/Documentation/sysrq.txt
# for what other values do
#kernel.sysrq=1

#####
# Protected links
#
# Protects against creating or following links under certain conditions
# Debian kernels have both set to 1 (restricted)
# See https://www.kernel.org/doc/Documentation/sysctl/fs.txt
#fs.protected_hardlinks=0
#fs.protected_symlinks=0

net.ipv4.ip_forward=1
```

C:\Users\Redmi\Desktop\Текстовый документ.txt

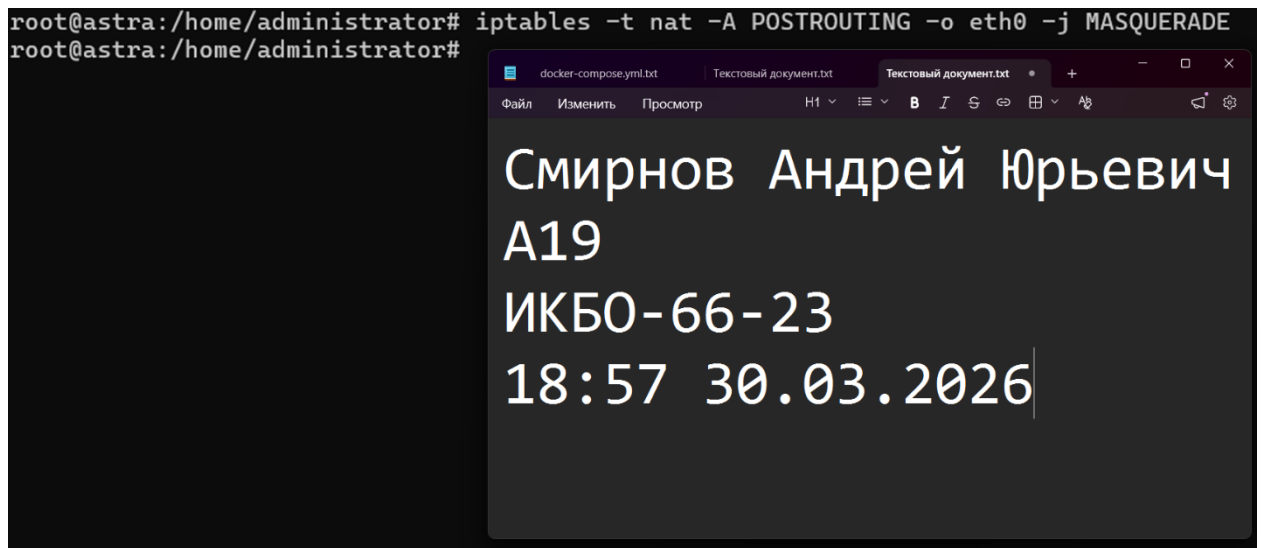
docker-compose.yml Текстовый докумен Текстовый док * + - □ ×

Файл Изменить Просмотр H1 B I S ...

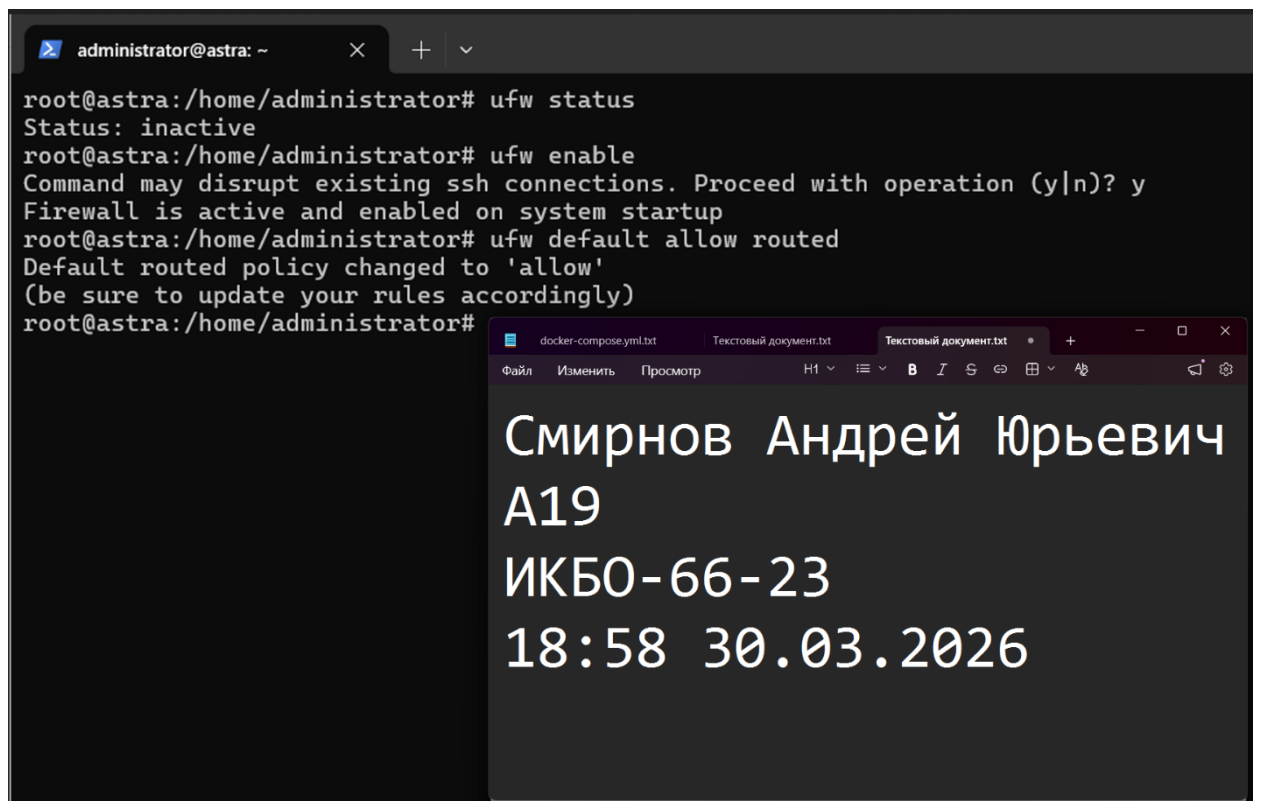
Смирнов Андрей
Юрьевич A19
ИКБО-66-23
18:55 30.03.2026

7. Включение маскардинга

```
root@astra:/home/administrator# iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
root@astra:/home/administrator#
```

A screenshot of a terminal window. The terminal shows the execution of the command `iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE`. A semi-transparent text box is overlaid on the right side of the terminal, containing the text: "Смирнов Андрей Юрьевич", "A19", "ИКБО-66-23", and "18:57 30.03.2026". The text box has a dark background and white text.

8. Работа с ufw

A screenshot of a terminal window showing the configuration of ufw. The terminal output is as follows:

```
administrator@astra: ~
root@astra:/home/administrator# ufw status
Status: inactive
root@astra:/home/administrator# ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@astra:/home/administrator# ufw default allow routed
Default routed policy changed to 'allow'
(be sure to update your rules accordingly)
root@astra:/home/administrator#
```

A semi-transparent text box is overlaid on the right side of the terminal, containing the text: "Смирнов Андрей Юрьевич", "A19", "ИКБО-66-23", and "18:58 30.03.2026". The text box has a dark background and white text.

9. Конфигурация /etc/ufw/before.rules

```
GNU nano 2.7.4 Файл: /etc/ufw/before.rules Изменён
-A ufw-not-local -j DROP

# allow MULTICAST mDNS for service discovery (be sure the MULTICAST line above
# is uncommented)
-A ufw-before-input -p udp -d 224.0.0.251 --dport 5353 -j ACCEPT

# allow MULTICAST UPnP for service discovery (be sure the MULTICAST line above
# is uncommented)
-A ufw-before-input -p udp -d 239.255.255.250 --dport 1900 -j ACCEPT

# don't delete the 'COMMIT' line or these rules won't be processed
COMMIT

*nat
:POSTROUTING ACCEPT [0:0]
#Forwardtraffic from eth1 through eth0.
-A POSTROUTING -s 192.168.1.0/24 -o eth0 -j MASQUERADE
#don't delete the 'COMMIT' line or these nat table rules won't
#be processed
COMMIT

^G Помощь ^O Записать ^W Поиск ^K Вырезать ^J Вывод
^X Выход ^R ЧитФайл ^N Замена ^U Отмен. вырез ^T Словарь ^K К строке ^V СледСтр ^-/ ПостСтрока

administrator@astra: ~
root@astra:/home/administrator# nano /etc/ufw/before.rules
root@astra:/home/administrator# ufw disable
Firewall stopped and disabled on system startup
root@astra:/home/administrator# ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
root@astra:/home/administrator#
```

Смирнов Андрей Юрьевич
A19
ИКБ0-66-23
19:00 30.03.2026

10. Вывод iptables -t nat -L


```
administrator@astra: ~
root@astra:/home/administrator# iptables -t nat -L
Chain PREROUTING (policy ACCEPT)
target     prot opt source                destination

Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination

Chain POSTROUTING (policy ACCEPT)
target     prot opt source                destination
MASQUERADE all  -- anywhere             anywhere
MASQUERADE all  -- 192.168.1.0/24      anywhere
root@astra:/home/administrator#
```

Смирнов Андрей Юрьевич
A19
ИКБО-66-23
19:00 30.03.2026

11. Настройка интерфейса eth1 (файл конфигурации)

```
administrator@astra: ~
GNU nano 2.7.4 Файл: /etc/network/interfaces

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback
auto eth0
iface eth0 inet static
    address 10.0.99.15
    netmask 255.255.255.0
    gateway 10.0.99.1
    dns-nameservers 8.8.8.8

auto eth1
iface eth1 inet static
    address 192.168.1.1
    netmask 255.255.255.0
```

Смирнов Андрей
Юрьевич A19
ИКБО-66-23
19:02 30.03.2026

12. Отредактированный файл конфигурации /etc/dhcp/dhcpd.conf

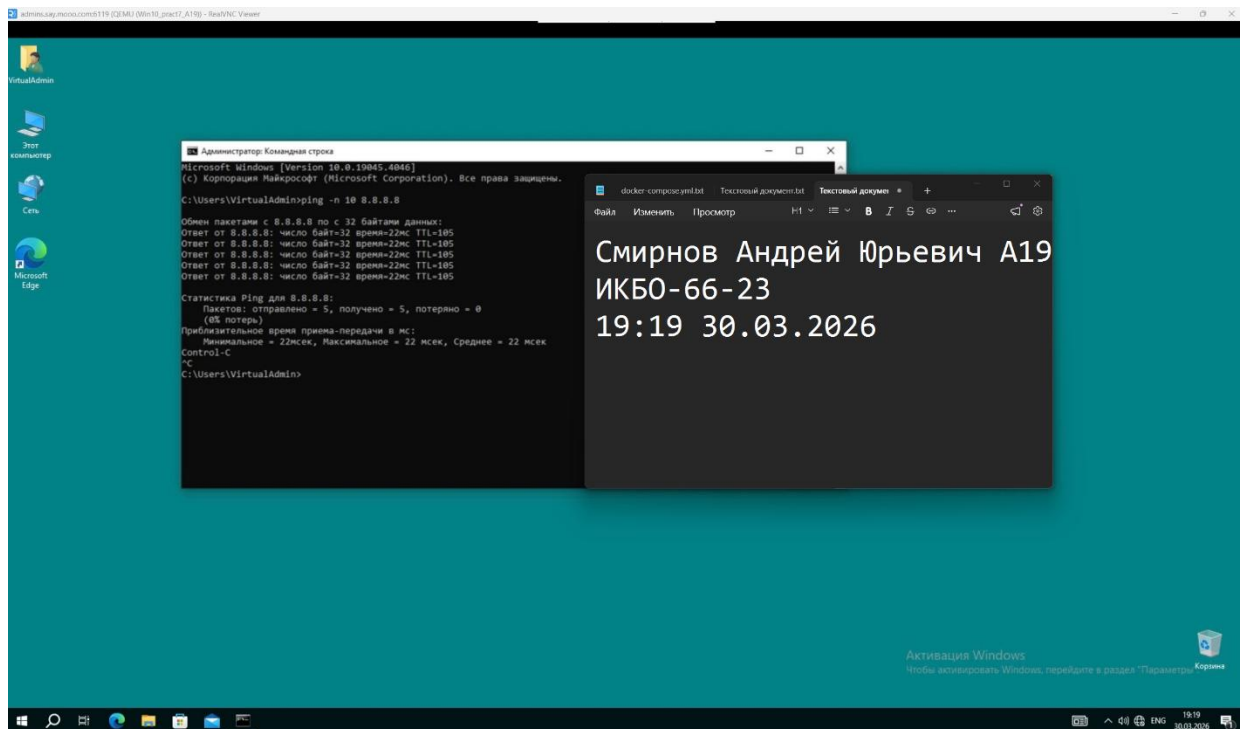
```
administrator@astra: ~
GNU nano 2.7.4 Файл: /etc/dhcp/dhcpd.conf

#       range 10.0.29.10 10.0.29.230;
#   }
#}

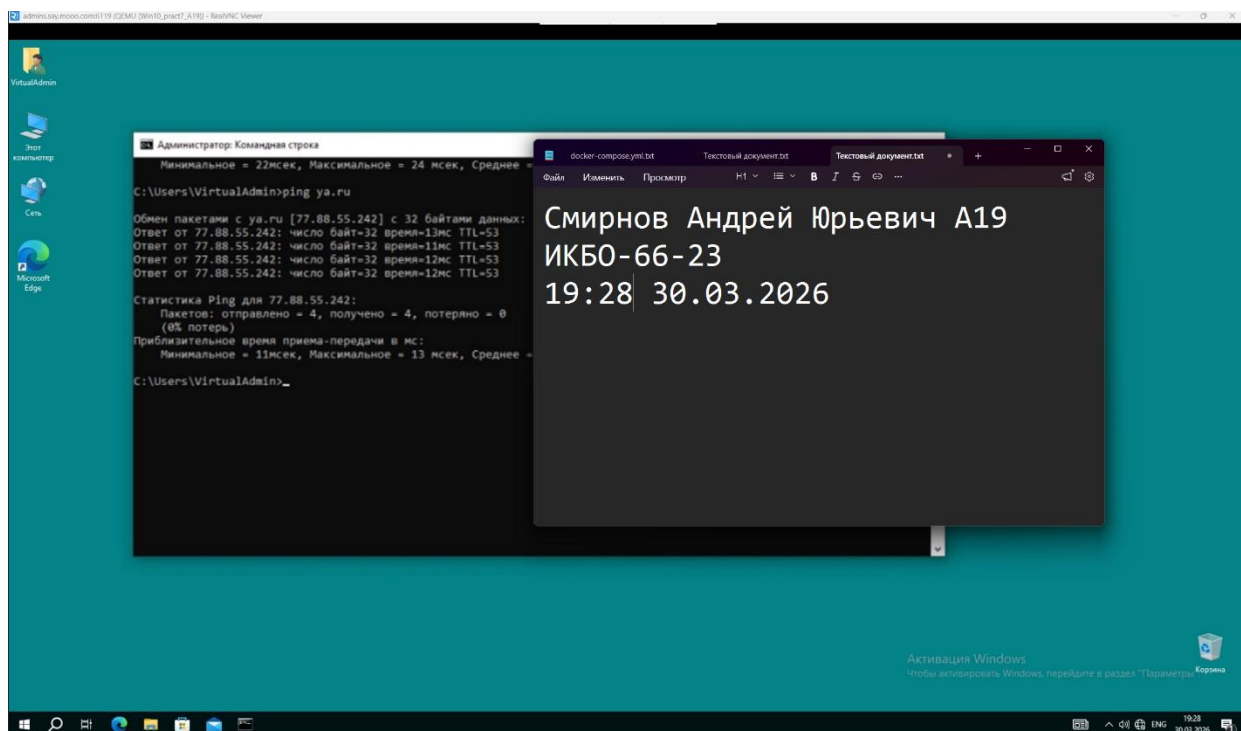
subnet 192.168.1.0 netmask 255.255.255.0 {
    range 192.168.1.50 192.168.1.240;
    option routers 192.168.1.1;
    option broadcast-address 192.168.1.255;
    option domain-name-servers 192.168.1.1;
    authoritative;
}
```

Смирнов Андрей Юрьевич
A19
ИКБО-66-23
19:04 30.03.2026

13. Проверка подключения к Интернету на VM с ОС Windows через ping 8.8.8.8



14. Проверка подключения к Интернету на VM с ОС Windows после настройки DNS через ping ya.ru



15. Подключение через PuTTY после настройки файрвола

